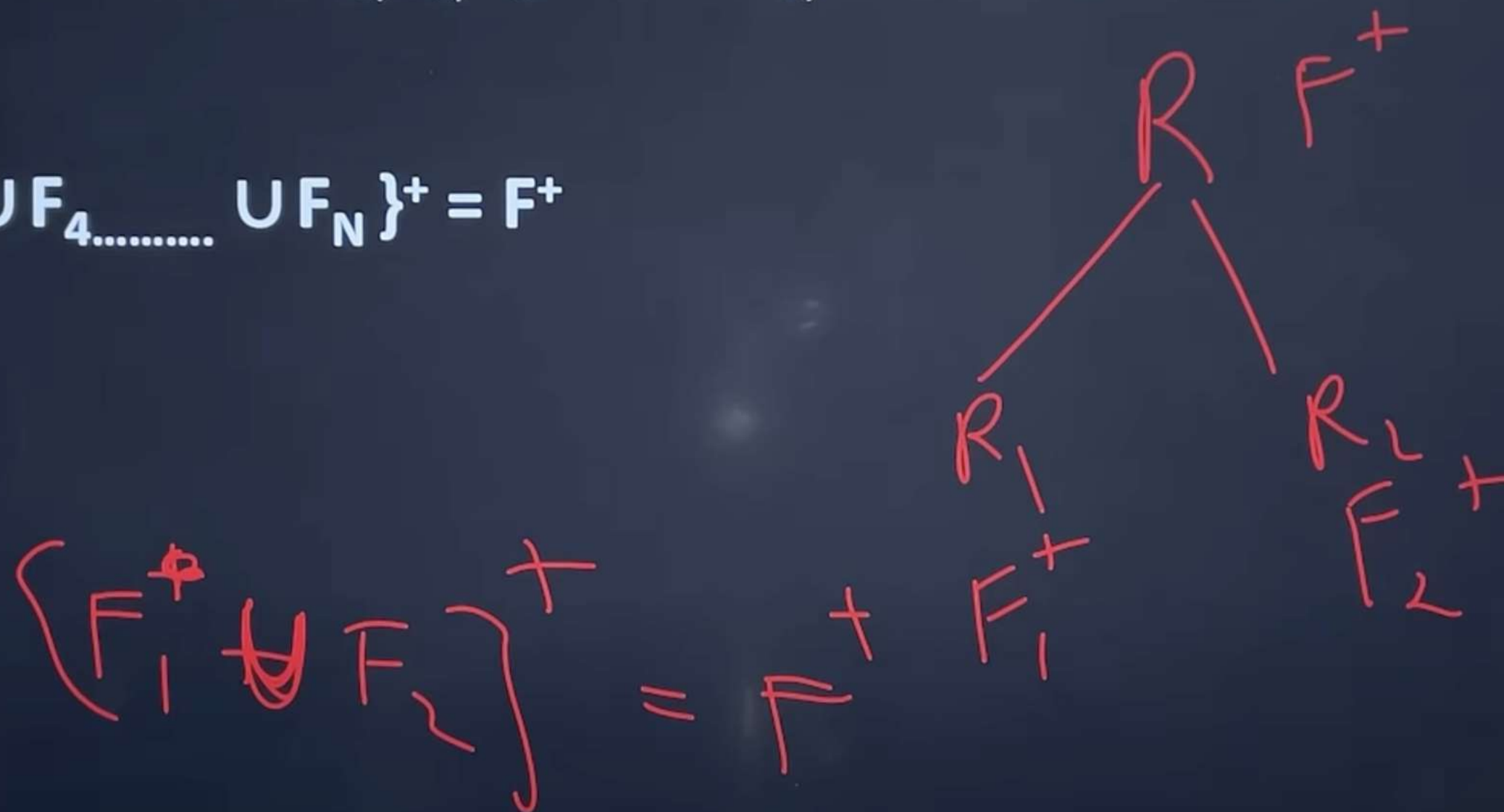


- Let relation R be decomposed into Relations $R_1, R_2, R_3, \dots, R_N$ with their respective functional Dependencies set as $F_1, F_2, F_3, \dots, F_N$, then the Decomposition is Dependency Preserving iff

- $\{F_1 \cup F_2 \cup F_3 \cup F_4 \dots \cup F_N\}^+ = F^+$



- Dependency preservation property, although desirable, is sometimes sacrificed

5 NF/Project-Join Normal Form

- A Relational table R is said to be in 5th normal form if
 - it is in 4 NF ✓
 - it cannot be further non-loss decomposed ✓



How to check for lossless join decomposition using FD set, following conditions must hold:

- Union of Attributes of R_1 and R_2 must be equal to attribute of R . Each attribute of R must be either in R_1 or in R_2 . $\text{Att}(R_1) \cup \text{Att}(R_2) = \text{Att}(R)$
- Intersection of Attributes of R_1 and R_2 must not be NULL. $\text{Att}(R_1) \cap \text{Att}(R_2) \neq \Phi$
- Common attribute must be a key for at least one relation (R_1 or R_2)
 - $\text{Att}(R_1) \cap \text{Att}(R_2) \rightarrow (R_1)$ or $\text{Att}(R_1) \cap \text{Att}(R_2) \rightarrow (R_2)$

R

A	B	C	D	E
A	122	1	W	A
E	236	4	X	B
A	199	1	Y	C
B	213	2	Z	D

① $R_1(A, B) \quad R_2(C, D)$

$R_1(A, \underline{B}) \quad R_2(\underline{D}, \underline{E})$

② $R_1(ABC) \quad R_2(D, E)$

③ $R_1(ABC) \quad R_2(C, D, E)$

Chapter-4(Normalization)

Lossless Decomposition

- Decomposition is lossless if $R_1 \bowtie R_2 = R$ "The decomposition of relation R into R_1 and R_2 is lossless when the join of R_1 and R_2 yield the same relation as in R." which guarantees that the spurious (extra or less) tuple generation problem does not occur with respect to the relation schemas created after decomposition.

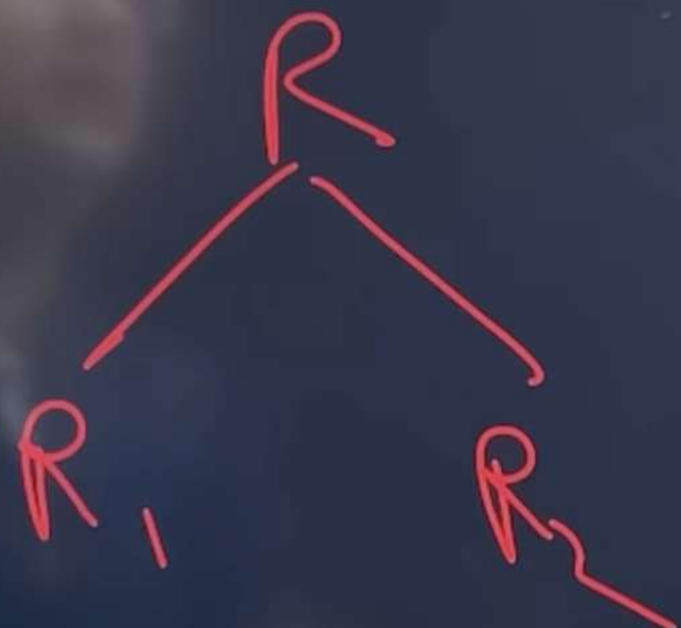
- This property is extremely critical and must be achieved at any c

- lossless Decomposition / NonAdditive Join Decomposition



- Decomposition is lossy if $R_1 \bowtie R_2 \supset R$

Decomposition is lossy if $R \supset R_1 \bowtie R_2$

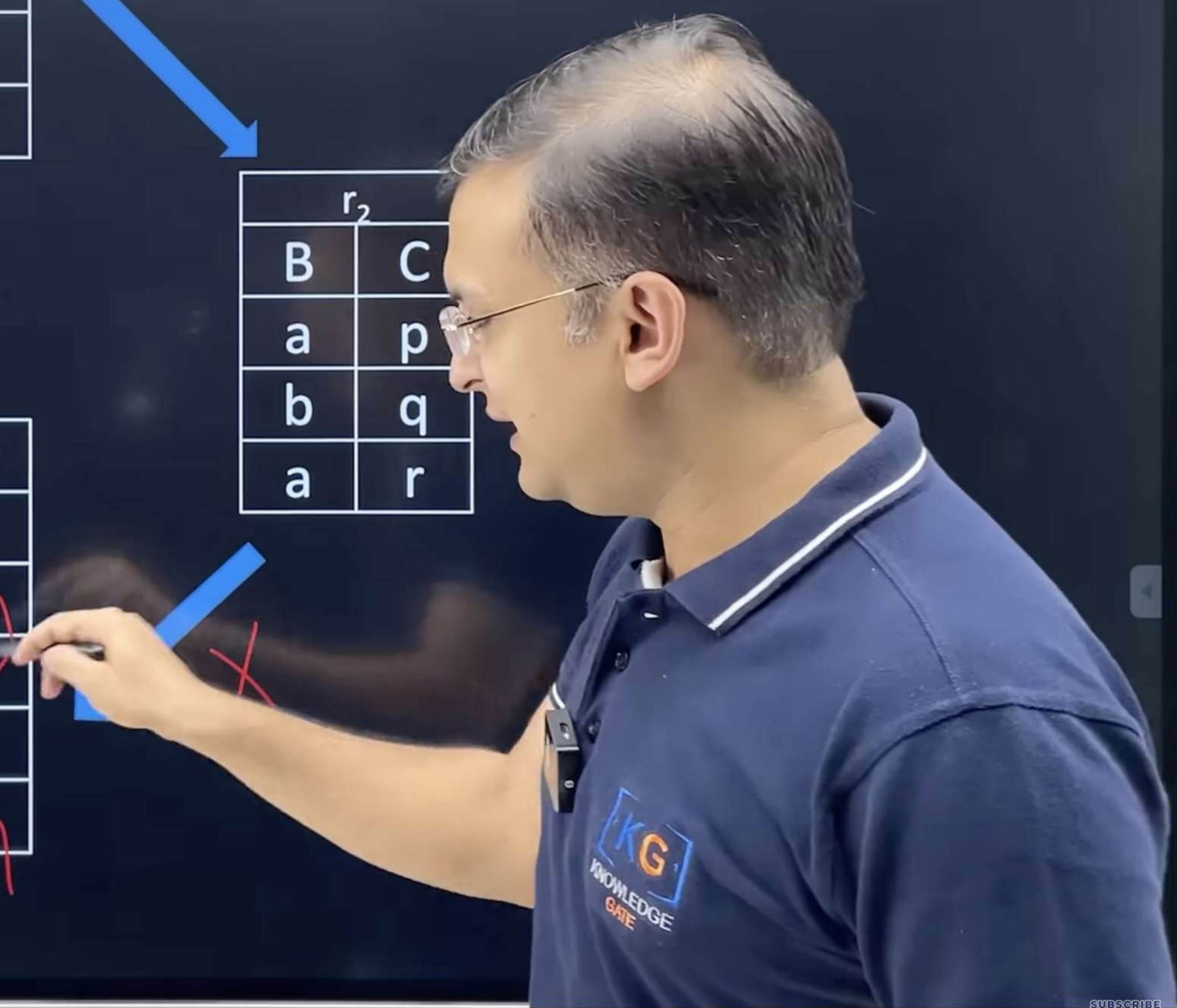


r ₁	
A	B
1	a
2	b
3	a

r		
A	B	C
1	a	p
2	b	q
3	a	r

r ₂	
B	C
a	p
b	q
a	r

A	B	C
1	a	p
1	a	q
2	b	q
3	a	p
3	a	q



- if we decompose a table r into two tables r_1 and r_2 because of normalization then at some later stage if we want to join(combine) (natural join) these tables r_1 and r_2 , then we must get back the original table r , without any extra or less tuple. But some information may be lost during retrieval of original relation or table.

For e.g.

r		
A	B	C
1	a	p
2	b	q
3	a	r

r ₁	
A	B
1	a
2	b
3	a

r ₂	
B	C
a	p
b	q
a	r

Lossy/Lossless-Dependency Preserving Decomposition

- Because of a normalization a table is Decomposed into two or more tables, but during this decomposition we must ensure satisfaction of some properties out of which the most important is lossless join property / decomposition.



Chapter-4(Normalization)

4NF

- A relation is in 4NF iff

- It is in BCNF ✓

- There must not exist any non-trivial multivalued dependency.

- Each MVD is decomposed in separate table, where it becomes trivial MVD



Chapter-4(Normalization)

- Each row indicates that a given restaurant can deliver a given variety. The table has no non-key attributes because its only key is {Restaurant, Variety, Delivery Area}. Therefore, it meets all normal forms up to BCNF.

- If we assume, however, that Variety offered by a restaurant are not affected by delivery area (i.e. a restaurant offers all Variety it makes to all areas it supplies), then it does not meet 4NF. The problem is that the table features two non-trivial multivalued dependencies on the {Restaurant} attribute (which is not a super key). The dependencies are:

- {Restaurant} $\twoheadrightarrow$ {Variety}
- {Restaurant} $\twoheadrightarrow$ {Delivery Area}

Multivalued Dependency

Restaurant Delivery Permutations		
Restaurant	Variety	Delivery Area
Chatora Sweets	Samosa	Hatibagan Market
Chatora Sweets	Samosa	Chandni Chowk
Chatora Sweets	Samosa	Koramangala
Chatora Sweets	Dosa	Hatibagan Market
Chatora Sweets	Dosa	Chandni Chowk
Chatora Sweets	Dosa	Koramangala
Moolchand	Ladoo	Koramangala
Moolchand	Dosa	Koramangala
Thaggu	Samosa	Hatibagan Market
Thaggu	Samosa	Chandni Chowk
Thaggu	Ladoo	Hatibagan Market
Thaggu	Ladoo	Chandni Chowk

E.g. let the constraint specified by MVD in relation **Student** as

$S_name \twoheadrightarrow Club_name$

$S_name \twoheadrightarrow P_no$

S_Name	Club_name
Kamesh	Dance
Kamesh	Guitar

S_Name	P_no
Kamesh	123
Kamesh	789

S_Name	Club_name	P_no
Kamesh	Dance	123
Kamesh	Guitar	123
Kamesh	Dance	789
Kamesh	Guitar	789

NOTE: The above Student schema is in BCNF as no functional dependency holds on EMP, but still redundancy due to MVD.

- A **trivial multivalued dependency** $X \twoheadrightarrow Y$ is one where either Y is a subset of X , or X and Y together form the whole set of attributes of the relation.

 $S_name \twoheadrightarrow Club_name$

S_Name	Club_name
Kamesh	Dance
Kamesh	Guitar

 $S_name \twoheadrightarrow Club_name$
 $S_name \twoheadrightarrow P_no$

S_Name	Club_name	P_no
Kamesh	Dance	123
Kamesh	Guitar	123
Kamesh	Dance	789
Kamesh	Guitar	789

Multivalued Dependency

- Denoted by, $A \twoheadrightarrow B$, Means, for every value of A, there may exist more than one value of B.

• E.g. $S_name \twoheadrightarrow Club_name$

S_Name	Club_name
Kamesh	Dance
Kamesh	Guitar

BCNF (BOYCE CODD NORMAL FORM)

- A relational schema R is said to be BCNF if every functional dependency in R from

- $\alpha \rightarrow \beta$
- α must be a super key ✓



THIRD NORMAL FORM DIRECT DEFINATION

- A relational schema R is said to be 3 NF if every functional dependency in R from $\alpha \rightarrow \beta$, either α is super key or β is the prime attri

$$\alpha \rightarrow \beta$$

$$\alpha \rightarrow \beta$$

P.D.

$$P \rightarrow NP$$

T.D.

$$NP \rightarrow NP$$

P

THIRD NORMAL FORM

- Let R be the relational schema, it is said to be in 3 NF
 - R should be in 2NF ✓
 - It must not contain any transitive dependency



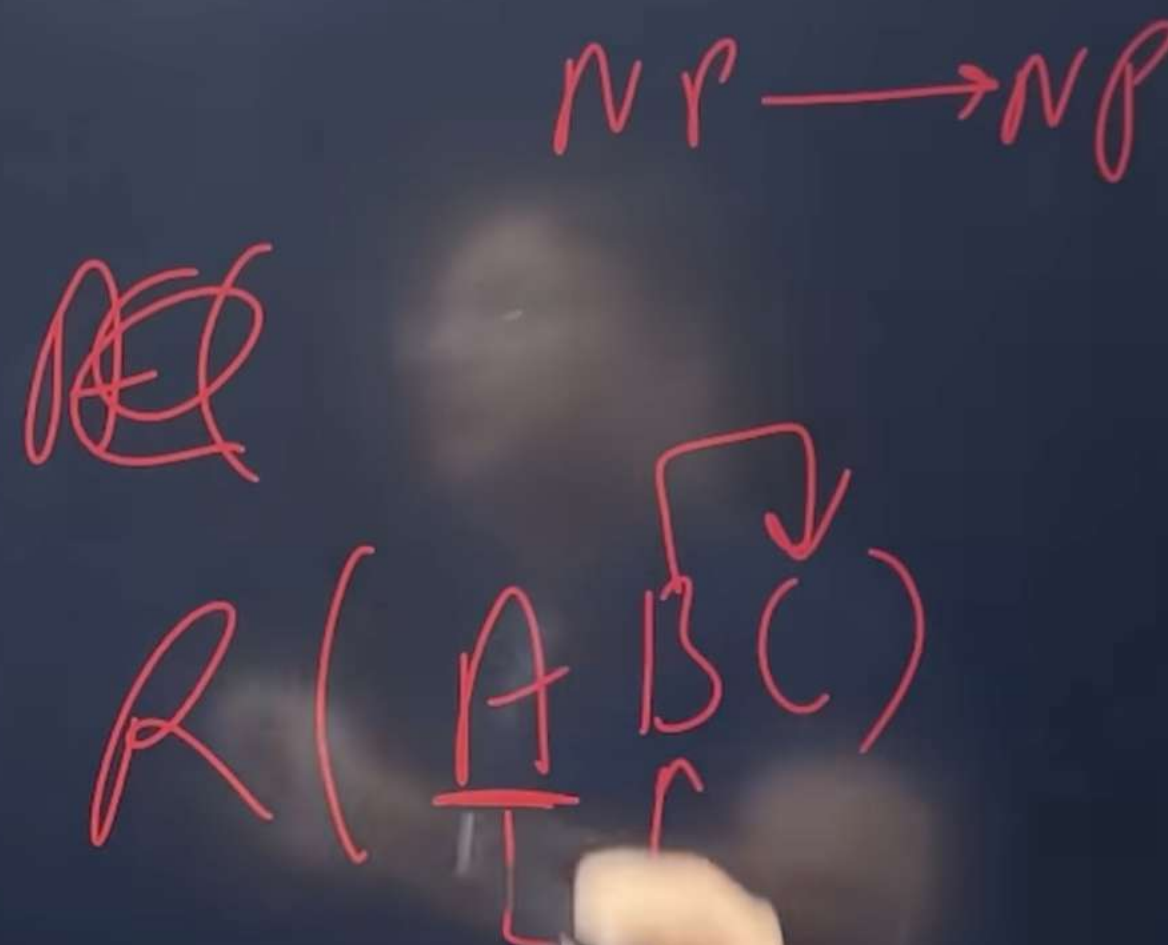
TRANSITIVE DEPENDENCY – A functional dependency from non-Prime attribute to non-Prime attribute is called transitive

E.g.- R(A, B, C, D) with A as a candidate key

$A \rightarrow B$

$B \rightarrow C$ [transitive dependency]

$A \rightarrow D$ [transitive dependency]



Chapter-4(Normalization)

 $QR(A, B, C) \quad B \rightarrow C$

A	B	C
a	1	X
b	2	Y
a	3	Z
C	3	Z
D	3	Z
E	3	Z

A	B
A	1
B	2
A	3
C	3
D	3
E	3

B	C
1	X
2	Y
3	Z



SECOND NORMAL FORM

- Relation R is in 2NF if,
 - R should be in 1 NF.
 - R should not contain any Partial dependency. (that is every non-prime attribute should be fully dependent upon candidate key)

